

T1036.007 Test Number 1 Masquerading: Double File Extension

1. Executive Summary

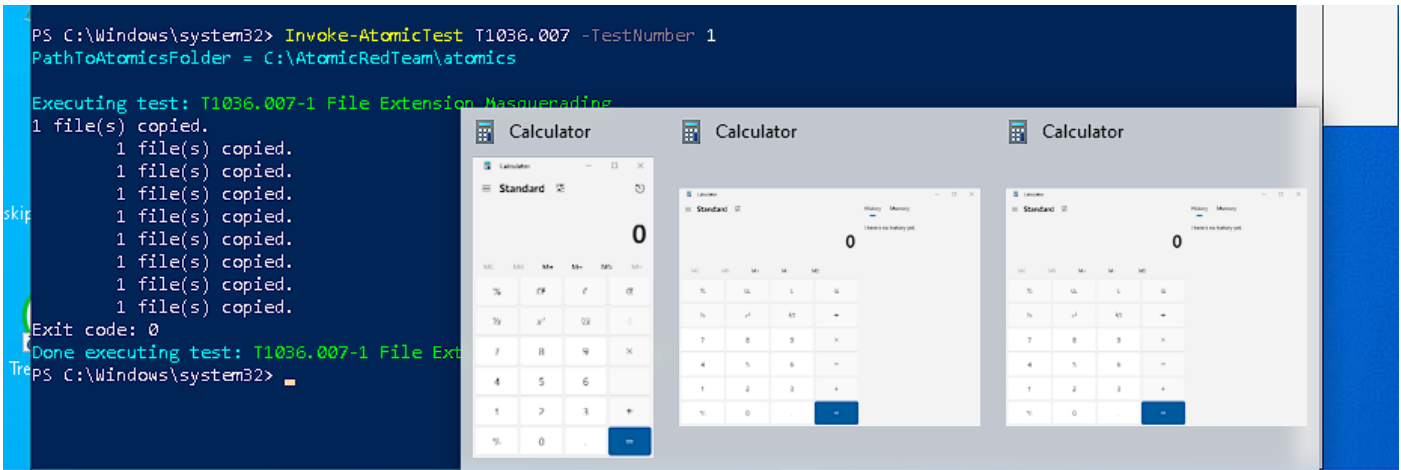
Bài kiểm thử Atomic Red Team T1036.007-1 - File Extension Masquerading được thực hiện trên máy Win10-Victim nhằm mô phỏng kỹ thuật dùng phần mở rộng kép để làm file thực thi hoặc script trông giống tài liệu, hình ảnh hợp lệ.

Bài test tạo và thực thi 9 file nguy trang trong thư mục %TEMP%, gồm:

- Ba file thực thi: .docx.exe, .pdf.exe, .ps1.exe.
- Ba VBScript: .xls.vbs, .xlsx.vbs, .png.vbs.
- Ba PowerShell script: .doc.ps1, .pdf.ps1, .rtf.ps1.

LimaCharlie phát hiện chuỗi hành vi, Splunk tổng hợp thành cảnh báo Critical, Risk Score 90, sau đó chuyển sang DFIR-IRIS.

Kết luận: True Positive - Authorized Simulation. Hành vi nguy trang và thực thi file đã xảy ra thành công. Dù tên một D&R rule có chứa từ “Block”, bằng chứng cho thấy các payload thử nghiệm vẫn được thực thi, nên khả năng ngăn chặn cần được kiểm tra lại.



Kiểm tra trong thư mục %TEMP% thì các file đã được copy thành công

T1036.007_masquerading.doc	6/24/2026 3:32 PM	Windows PowerS...
T1036.007_masquerading.docx	12/7/2019 4:09 PM	Application
T1036.007_masquerading.pdf	12/7/2019 4:09 PM	Application
T1036.007_masquerading.pdf	6/24/2026 3:32 PM	Windows PowerS...
T1036.007_masquerading.png	6/24/2026 3:32 PM	VBScript Script File
T1036.007_masquerading.ps1	12/7/2019 4:09 PM	Application
T1036.007_masquerading.rtf	6/24/2026 3:32 PM	Windows PowerS...
T1036.007_masquerading.xls	6/24/2026 3:32 PM	VBScript Script File
T1036.007_masquerading.xlsx	6/24/2026 3:32 PM	VBScript Script File
tmp45CE.tmp	7/27/2026 3:30 PM	TMP File

2. Incident Classification

Hạng mục	Kết quả
Case Name	T1036.007-1 - File Extension Masquerading
Classification	True Positive - Authorized Simulation
MITRE Tactic	Stealth
MITRE Technique	T1036.007 - Double File Extension
Atomic Test	Test #1 - File Extension Masquerading

Affected Host	Win10-Victim / DESKTOP-JI0TPU0
Executing User	DESKTOP-JI0TPU0\longday
Excution Location	%TEMP% - User-writable directory
Alert Severity	Critical
Risk Score	90
Business Impact	None - Controlled Lab
Case Status	Ready for Closure after cleanup verification

3. Test Objective and Scope

Bài kiểm thử nhằm:

- Mô phỏng file nguy hiểm mang tên giống tài liệu hoặc hình ảnh.
- Xác nhận khả năng theo dõi file được tạo trong thư mục người dùng có quyền ghi.
- Kiểm tra khả năng phát hiện thực thi .exe, .vbs và .ps1 có phần mở rộng kép.
- Đánh giá quá trình LimaCharlie → Splunk → DFIR-IRIS.
- Xác minh D&R rule có thực sự ngăn chặn hay chỉ phát hiện hành vi.

Phạm vi giới hạn trên một máy Windows 10 victim trong Super SOC HomeLab.

4. Detection Summary

Chuỗi lệnh sử dụng cmd.exe để sao chép calc.exe cùng các script Atomic Red Team sang %TEMP%, đổi tên thành 9 file có phần mở rộng kép rồi thực thi lần lượt.

```
16:35:12 Non Interactive PowerShell Process Spawned → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140702580146176,"COMMAND_LINE":"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -File C:\\Users\\longday\\AppData\\Local\\Temp
16:35:12 LC - Ransomware VIP L2 Suspicious Filename Executed Block → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140702580146176,"COMMAND_LINE":"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -File C:\\Users\\longday\\AppDa
16:35:12 Suspicious Execution of Hostname → desktop-j10tpu0 ["event":{"COMMAND_LINE":"","C:\\Windows\\System32\\HOSTNAME.EXE","", "FILE_IS_SIGNED":1, "FILE_PATH":"C:\\Windows\\System32\\HOSTNAME.EXE", "HASH":"a90c3fb350a11c6f0a6defa9607987d92
16:35:12 Suspicious Execution of Hostname → desktop-j10tpu0 ["event":{"COMMAND_LINE":"","C:\\Windows\\System32\\HOSTNAME.EXE","", "FILE_IS_SIGNED":1, "FILE_PATH":"C:\\Windows\\System32\\HOSTNAME.EXE", "HASH":"a90c3fb350a11c6f0a6defa9607987d92
16:35:10 LC - Ransomware VIP L1 User Writable Suspicious Execution → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140699060998928,"COMMAND_LINE":"C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.png.vbs", "FILE_I
16:35:10 LC - Ransomware VIP L2 Suspicious Filename Executed Block → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140702580146176,"COMMAND_LINE":"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -File C:\\Users\\longday\\AppDa
16:35:10 Non Interactive PowerShell Process Spawned → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140702580146176,"COMMAND_LINE":"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -File C:\\Users\\longday\\AppData\\Local\\Temp
16:35:10 Potentially Suspicious PowerShell Script Execution From Temp Folder → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140702580146176,"COMMAND_LINE":"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -File C:\\Users\\long
16:35:10 Script Interpreter Execution From Suspicious Folder → desktop-j10tpu0 ["event":{"COMMAND_LINE":"","C:\\Windows\\System32\\WScript.exe", "C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.png.vbs", "FILE_IS_SIG
16:35:10 Potential Dropper Script Execution Via WScript/CScript/MSHTA → desktop-j10tpu0 ["event":{"COMMAND_LINE":"","C:\\Windows\\System32\\WScript.exe", "C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.png.vbs", "FI
16:35:10 Cscript/Uncommon Script Extension Execution → desktop-j10tpu0 ["event":{"COMMAND_LINE":"","C:\\Windows\\System32\\WScript.exe", "C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.png.vbs", "FILE_IS_SIGN
16:35:10 LC - Ransomware VIP L1 User Writable Suspicious Execution → desktop-j10tpu0 ["event":{"COMMAND_LINE":"","C:\\Windows\\System32\\WScript.exe", "C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.png.vbs", "FILE_I
16:35:10 LC - Ransomware VIP L2 Suspicious Filename Executed Block → desktop-j10tpu0 ["event":{"COMMAND_LINE":"","C:\\Windows\\System32\\WScript.exe", "C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.png.vbs", "FILE_I
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16:35:05 LC - Suspicious Process from Temp Directory → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140699060998928,"COMMAND_LINE":"C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.ps1.exe", "FILE_IS_SIGNED":1, "FILE_PATH":"C
16:35:03 LC - Ransomware VIP L2 Suspicious Filename Executed Block → desktop-j10tpu0 ["event":{"BASE_ADDRESS":14069812153488,"COMMAND_LINE":"C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.pdf.exe", "FILE_IS_SIGNED":1, "FILE_PATH":"C
16:35:03 Suspicious Double Extension File Execution → desktop-j10tpu0 ["event":{"BASE_ADDRESS":14069812153488,"COMMAND_LINE":"C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.pdf.exe", "FILE_IS_SIGNED":1, "FILE_PATH":"C
16:35:03 LC - Suspicious Process from Temp Directory → desktop-j10tpu0 ["event":{"BASE_ADDRESS":14069812153488,"COMMAND_LINE":"C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.pdf.exe", "FILE_IS_SIGNED":1, "FILE_PATH":"C
16:35:01 Suspicious Double Extension File Execution → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140695090427456,"COMMAND_LINE":"C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.docx.exe", "FILE_IS_SIGNED":1, "FILE_PATH":"C
16:35:01 LC - Ransomware VIP L2 Suspicious Filename Executed Block → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140695090427456,"COMMAND_LINE":"C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.docx.exe", "FILE_IS_SIGNED":1
16:35:01 LC - Suspicious Process from Temp Directory → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140695090427456,"COMMAND_LINE":"C:\\Users\\longday\\AppData\\Local\\Temp\\T1036.007_masquerading.docx.exe", "FILE_IS_SIGNED":1, "FILE_PATH":"C
16:34:50 LOL-Binary Copied From System Directory → desktop-j10tpu0 ["event":{"BASE_ADDRESS":140695090427456,"COMMAND_LINE":"cmd.exe /c copy C:\\Windows\\System32\\cmd.exe \\Temp\\T1036.007_masquerading.docx.exe /Y & copy C:\\
16:34:50 Local Accounts Discovery → desktop-j10tpu0 ["event":{"COMMAND_LINE":"","C:\\Windows\\System32\\whoami.exe", "FILE_IS_SIGNED":1, "FILE_PATH":"C:\\Windows\\System32\\whoami.exe", "HASH":"2d402a0d4d49e8c8bf7080e463155958fec397440438
16:34:50 Suspicious Execution of Hostname → desktop-j10tpu0 ["event":{"COMMAND_LINE":"","C:\\Windows\\System32\\HOSTNAME.EXE","", "FILE_IS_SIGNED":1, "FILE_PATH":"C:\\Windows\\System32\\HOSTNAME.EXE", "HASH":"a90c3fb350a11c6f0a6defa9607987d92
```

Các detection nổi bật gồm:

- Suspicious Double Extension File Execution
- LOL-Binary Copied From System Directory
- LC - Suspicious Process from Temp Directory
- Cscript/Wscript Uncommon Script Extension Execution
- Potential Dropper Script Execution Via WScript/CScript/MSHTA
- Script Interpreter Execution From Suspicious Folder
- Potentially Suspicious PowerShell Script Execution From Temp Folder
- LC - Ransomware VIP L1 User Writable Suspicious Execution
- LC - Ransomware VIP L2 Suspicious Filename Executed Block

Splunk hợp nhất chuỗi detection thành một cảnh báo DFIR-IRIS:

- LimaCharlie EDR [CRITICAL]
- LC - Ransomware VIP L1 User Writable Suspicious Execution
- Risk Score: 90

LimaCharlie EDR - SOC Operations Center

Real-time EDR monitoring dashboard for LimaCharlie detections with MITRE ATT&CK mapping, endpoint analysis, and suspicious process tracking.

Time Ran...

Last 15 minutes

Total Detections

44

Unique Endpoints

1

High/Critical Alerts

22

MITRE Techniques

9

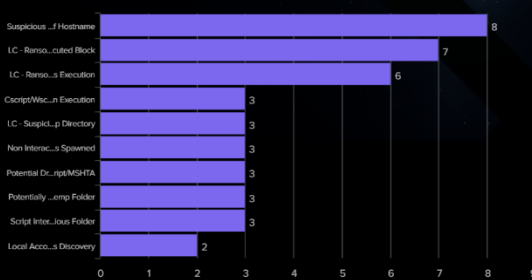
Active D&R Rules

12

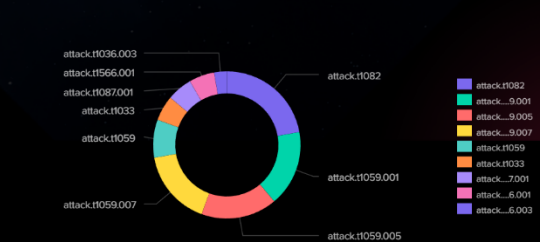
Detection Timeline



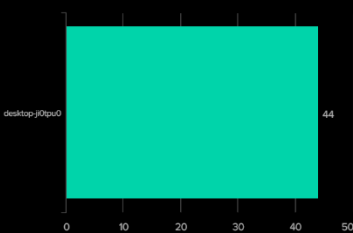
Top Detection Categories



MITRE ATT&CK Techniques



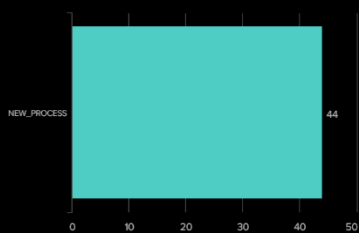
Top Affected Endpoints



Severity Breakdown



Detection by Event Type



Recent Detections

_time	Severity	Detecti...	Endpoi...	Source...	External...	Event_Ty...	MITRE	Rule	Auth...
2026-07-27T16:35:18.000+07:00	MEDIUM	Potentially Suspicious Powershell Script Execution From Temp Folder	desktop-j10tpu0	172.16.1.0	123.24.225.221	NEW_PROCES S	attack.t1059.001	service.window_s_process_creation/proc_creation_win_powershell_script_execution_from_temp_folder	_ext-sigma-7a14fbc3-54d9-4b4d-8700-61edddada04f0[bulk][segment]
2026-07-27T16:35:18.000+07:00	LOW	Suspicious Execution of Hostname	desktop-j10tpu0	172.16.1.0	123.24.225.221	NEW_PROCES S	attack.t1082	service.window_s_process_creation/proc_creation_win_hostname_execution	_ext-sigma-7a14fbc3-54d9-4b4d-8700-61edddada04f0[bulk][segment]
2026-07-27T16:35:18.000	LOW	Suspicious Execution of	desktop-j10tpu0	172.16.1.0	123.24.225.221	NEW_PROCES S	attack.t1082	service.window_s_process_creation/proc_creation_win_hostname_execution	_ext-sigma-7a14fbc3-54d9-4b4d-8700-61edddada04f0[bulk][segment]

< Prev 1 2 3 Next >

Suspicious Process Executions

_time	Endpoi...	User	Proce...	Command...	ParentPro...	PID	PPID	Hash	Detecti...
2026-07-27T16:35:18.000+07:00	desktop-j10tpu0	DESKTOP-J10TPU0\longday	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.rtf.ps1	C:\Windows\System32\cmd.exe	4784	6264	9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3	Potentially Suspicious Powershell Script Execution From Temp Folder
2026-07-27T16:35:18.000+07:00	desktop-j10tpu0	DESKTOP-J10TPU0\longday	C:\Windows\System32\HOSTNAME.EXE	"C:\Windows\System32\HOSTNAME.EXE"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	3892	4784	a90c3fb350a1fc616a6efa9607987d924d1de65e09ca9faf2e0e0e00531ee335	Suspicious Execution of Hostname

< Prev 1 2 Next >



LIMACHARLIE EDR [CRITICAL] LC - RANSOMWARE VIP L1 USER WRITABLE SUSPICIOUS EXECUTION ON WIN10-VICTIM

#131 - E9FCD5A4-4A4F-465C-A898-06A408170208

Merge

Assign

Set status

Close with note

Close

LimaCharlie detected a case-worthy endpoint event. Primary=LC - Ransomware VIP L1 User Writable Suspicious Execution; MatchedRules=Cscript/Wscript Uncommon Script Extension Execution | LC - Ransomware VIP L1 User Writable Suspicious Execution | LC - Ransomware VIP L2 Suspicious Filename Executed Block | Potential Dropper Script Execution Via WScript/CScript/MSHTA | Script Interpreter Execution From Suspicious Folder; Host=Win10-Victim; User=unknown; Process=wscript.exe; RiskScore=90; CommandLine="C:\Windows\System32\WScript.exe" "C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.xls.vbs"

General info

Source:

LimaCharlie EDR

Source Link:

http://u-splunk8000/app/supersoar_soc_app/search?q=%7Cloadjob%20scheduler_longday_c3vWzUzb2FyC3NvY19hcHA_RMD%5B46cb1d65f4889b9_at_1785145500_904%20%7C%20head%20%20%7C%20tail%201%20earliest=0&latest=now

Source Reference:

LC-EDR-V2-17f1e1bd3f8a1520e6d507112d481638

Source Event Time:

09:35:10 27/7/2026 UTC

IRIS Creation Time:

09:45:04 27/7/2026 UTC

Alert note

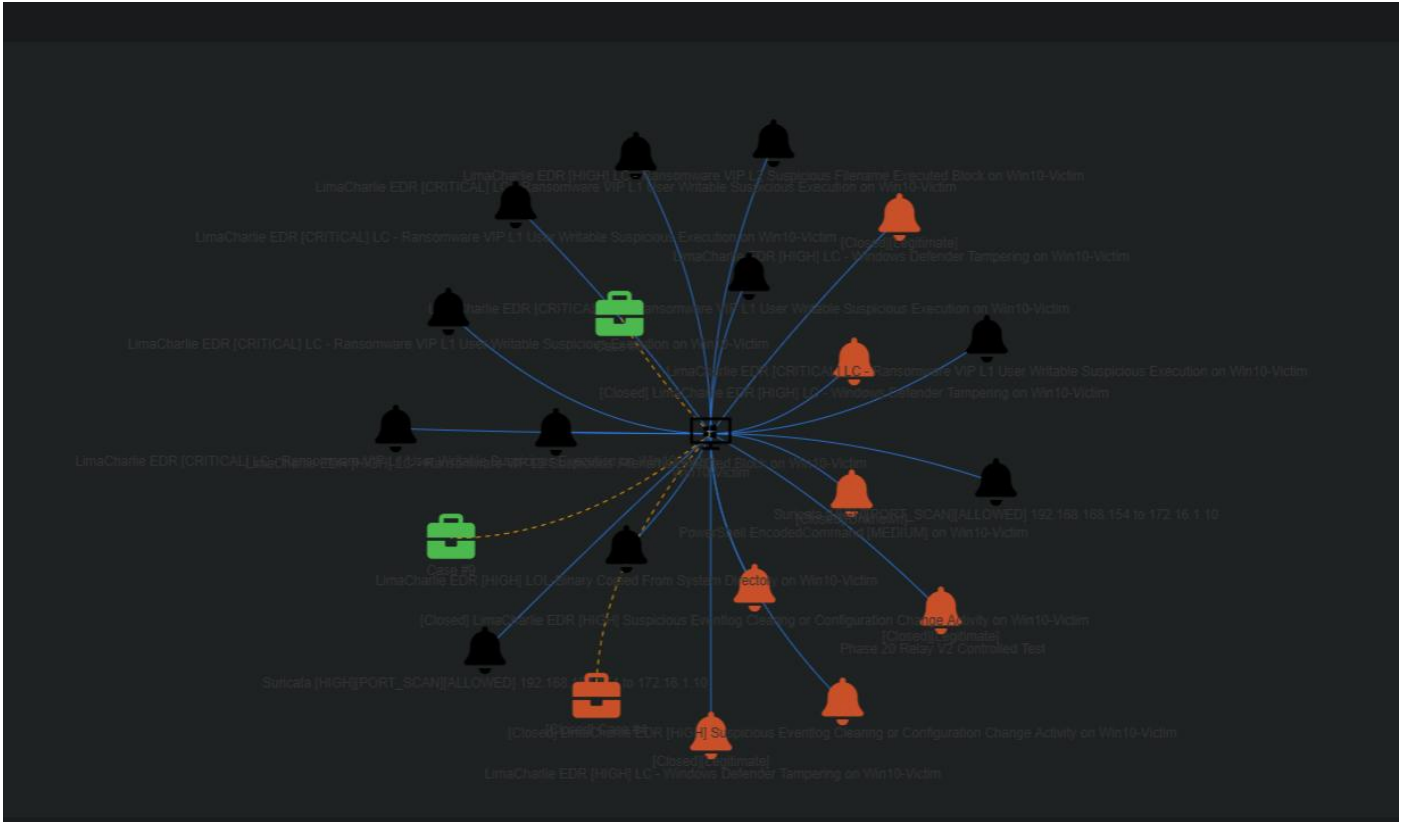
Trong alert trên DFIR-IRIS có kèm theo phần raw_event được splunk đẩy qua thì thấy chuỗi lệnh được match bởi sigma rule *LOL-Binary Copied From System Directory*

raw_event:

```
[{"author": "ext-sigma-7a14bc3-54d9-4b4d-8700-61edda04f0[bulk][segment]";cat:-:Potential Dropper Script Execution Via WScript/CScript/MSHTA";detect":
{"event":{"COMMAND_LINE":"C:\Windows\System32\WScript.exe" "C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.xls.vbs\"
"FILE_IS_SIGNED":1,"FILE_PATH":"C:\Windows\System32\WScript.exe","HASH":"40d9518bf166d1d94b4d60dfe40efb06d2dd92e1ca6d9ab1a4d2b8ccc49da532""P
ARENT":{"BASE_ADDRESS":"140695650500608","COMMAND_LINE":"cmd.exe" /c copy "C:\Windows\System32\calc.exe"
%temp%\T1036.007_masquerading.docx.exe /Y \u0026 copy "C:\Windows\System32\calc.exe" %temp%\T1036.007_masquerading.pdf.exe /Y \u0026 copy
"C:\Windows\System32\calc.exe" %temp%\T1036.007_masquerading.ps1.exe /Y \u0026 copy
"C:\AtomicRedTeam\atomic\T1036.007\src\T1036.007_masquerading.vbs" %temp%\T1036.007_masquerading.xls.vbs /Y \u0026 copy
"C:\AtomicRedTeam\atomic\T1036.007\src\T1036.007_masquerading.vbs" %temp%\T1036.007_masquerading.xlsx.vbs /Y \u0026 copy
"C:\AtomicRedTeam\atomic\T1036.007\src\T1036.007_masquerading.vbs" %temp%\T1036.007_masquerading.png.vbs /Y \u0026 copy
"C:\AtomicRedTeam\atomic\T1036.007\src\T1036.007_masquerading.ps1" %temp%\T1036.007_masquerading.doc.ps1 /Y \u0026 copy
"C:\AtomicRedTeam\atomic\T1036.007\src\T1036.007_masquerading.ps1" %temp%\T1036.007_masquerading.pdf.ps1 /Y \u0026 copy
"C:\AtomicRedTeam\atomic\T1036.007\src\T1036.007_masquerading.ps1" %temp%\T1036.007_masquerading.rtf.ps1 /Y \u0026
%temp%\T1036.007_masquerading.docx.exe \u0026 %temp%\T1036.007_masquerading.pdf.exe \u0026 %temp%\T1036.007_masquerading.ps1.exe \u0026
%temp%\T1036.007_masquerading.xls.vbs \u0026 %temp%\T1036.007_masquerading.xlsx.vbs \u0026 %temp%\T1036.007_masquerading.png.vbs \u0026
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.doc.ps1 \u0026
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.pdf.ps1 \u0026
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File
%temp%\T1036.007_masquerading.rtf.ps1";FILE_IS_SIGNED:1;"FILE_PATH":"C:\Windows\system32\cmd.exe";"HASH":"badf4752413cb0bdc03fb95820ca167f0cd
c63b597ccdb5ef4311180e088b0";"MEMORY_USAGE":4943872;"PARENT_ATOM":"55c2fa4c67cebdbe01d066f86a6708b1";"PARENT_PROCESS_ID":8908;"PROCES
S_ID":16147875;"TYPE":"LC-EDR-V2-17f1e1bd3f8a1520e6d507112d481638";"USER":"longday";"VENDOR":"LimaCharlie";"VULN":"CWE-200"}]
```

Giải mã các thành phần:

- Node trung tâm = 1 asset Win10-Victim, đây là điểm pivot của toàn bộ graph.
- Chuông cam = alert mức CRITICAL/HIGH liên quan đến asset này (VD: "Ransomware VIP L1 User Volatile Suspicious Execution", "Windows Defender Tampering")
- Chuông đen = alert mức thấp hơn (MEDIUM/LOW) hoặc đã được đóng/nhóm lại
- Icon vali xanh = các case đã được tạo/escalate từ nhóm alert đó (case #10, case #4...)
- Đường nét liền (xanh dương) = quan hệ alert → asset (alert này được ghi nhận trên asset trung tâm)
- Đường nét đứt (vàng cam) = quan hệ alert → case, tức những alert cụ thể đã được merge/escalate vào case đó thay vì đứng độc lập



Ý nghĩa thực tế của graph này:

Đây chính là bằng chứng trực quan cho thấy 1 asset (Win10-Victim) đang là tâm điểm của một chuỗi tấn công đa giai đoạn số lượng alert tỏa ra rất lớn (~20+ alert) từ nhiều rule khác nhau (Ransomware VIP, Defender Tampering, Suspicious Filename Executed Block, LOLBin Copied, Fortiscan ALLOWED...) đều hội tụ về cùng 1 máy, trong khung thời gian gần nhau.

5. Technical Timeline

Thời gian	Hoạt động	Kết quả
16:34:59	Atomic chạy HOSTNAME.exe thăm giờ môi trường. Chạy tiến trình whoami.exe xác minh quyền và user hiện tại sau đó chạy lệnh thực thi copy hàng loạt và thực thi các file vừa copy sang thư mục Temp	Trong khoảng 1s thì thực thi xong lệnh
16:35:01-16:35:19	Chạy .docx.exe, .pdf.exe và .ps1.exe	Ba phiên Calculator được khởi chạy
16:35:10-16:35:11	WScript.exe chạy .xls.vbs, .xlsx.vbs và .png.vbs	VBScript thực thi thành công
16:35:10-16:35:12	powershell.exe -File chạy .doc.ps1, .pdf.ps1 và .rtf.ps1	Các PowerShell script thực thi và gọi HOSTNAME.EXE
16:35:01	Thời điểm nguồn được ghi trong cảnh báo IRIS	LimaCharlie/Splunk ghi nhận chuỗi hành vi
16:45:01	Cảnh báo được tạo trên DFIR-IRIS	Pipeline hoàn tất sau khoảng 9 phút 54 giây

[illegible]

6. Host and SIEM Evidence

Các file chính được tạo tại

- C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.xls.vbs
- C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.xlsx.vbs
- C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.png.vbs

- C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.doc.ps1
- C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.pdf.ps1
- C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.rtf.ps1

Process telemetry xác nhận:

- Ba executable ngay trang khởi chạy Windows Calculator.
- WScript.exe thực thi ba VBScript.
- powershell.exe thực thi ba PowerShell script.
- HOSTNAME.EXE được gọi từ các PowerShell script.
- Splunk hiển thị đầy đủ Image, CommandLine, ParentImage và thời gian thực thi.

Time	Image	✓	CommandLine	✓	✓	ParentImage	✓
2026-07-27 16:35:16.623	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading_rtf.ps1		"2280"	C:\Windows\System32\cmd.exe	
2026-07-27 16:35:16.488	C:\Windows\System32\HOSTNAME.EXE		"C:\Windows\system32\HOSTNAME.EXE"		"2280"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
2026-07-27 16:35:16.407	C:\Windows\System32\HOSTNAME.EXE		"C:\Windows\system32\HOSTNAME.EXE"		"2280"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
2026-07-27 16:35:15.961	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading_pdf.ps1		"2280"	C:\Windows\System32\cmd.exe	
2026-07-27 16:35:15.738	C:\Windows\System32\HOSTNAME.EXE		"C:\Windows\system32\HOSTNAME.EXE"		"2280"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
2026-07-27 16:35:15.577	C:\Windows\System32\HOSTNAME.EXE		"C:\Windows\system32\HOSTNAME.EXE"		"2280"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
2026-07-27 16:35:13.865	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading_doc.ps1		"2280"	C:\Windows\System32\cmd.exe	
2026-07-27 16:35:13.886	C:\Windows\System32\wscript.exe		"C:\Windows\System32\WScript.exe"		"2280"	C:\Windows\System32\cmd.exe	
2026-07-27 16:35:13.542	C:\Windows\System32\wscript.exe		"C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading_xlsx.vbs"		"2280"	C:\Windows\System32\cmd.exe	
2026-07-27 16:35:13.170	C:\Windows\System32\wscript.exe		"C:\Windows\System32\WScript.exe"		"2280"	C:\Windows\System32\cmd.exe	
2026-07-27 16:35:11.457	C:\Windows\System32\smartscreen.exe		C:\Windows\System32\smartscreen.exe -Embedding		"2280"	C:\Windows\System32\svchost.exe	
2026-07-27 16:35:08.363	C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.ps1.exe		C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.ps1.exe		"2280"	C:\Windows\System32\cmd.exe	
2026-07-27 16:35:06.441	C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.pdf.exe		C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.pdf.exe		"2280"	C:\Windows\System32\cmd.exe	
2026-07-27 16:35:04.477	C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.docx.exe		C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading.docx.exe		"2280"	C:\Windows\System32\cmd.exe	
2026-07-27	C:\Windows\system32\cmd.exe				"2280"		

2026-07-27 16:35:03.897	C:\Windows\system32\cmd.exe				"2280"		
2026-07-27 16:35:03.891	C:\Windows\system32\cmd.exe				"2280"		
2026-07-27 16:35:03.849	C:\Windows\system32\cmd.exe				"2280"		
2026-07-27 16:35:03.842	C:\Windows\system32\cmd.exe				"2280"		
2026-07-27 16:35:03.839	C:\Windows\system32\cmd.exe				"2280"		
2026-07-27 16:35:02.889	C:\Windows\System32\cmd.exe		"cmd.exe" /c copy "C:\Windows\System32\calc.exe" %temp%\T1036.007_masquerading.docx.exe /Y & copy "C:\Windows\System32\calc.exe" %temp%\T1036.007_masquerading.pdf.exe /Y & copy "C:\Windows\System32\calc.exe" %temp%\T1036.007_masquerading.ps1.exe /Y & copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.vbs" %temp%\T1036.007_masquerading.xls.vbs /Y & copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.vbs" %temp%\T1036.007_masquerading.png.vbs /Y & copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.ps1" %temp%\T1036.007_masquerading.doc.ps1 /Y & copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.ps1" %temp%\T1036.007_masquerading.pdf.ps1 /Y & copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.ps1" %temp%\T1036.007_masquerading.rtf.ps1 /Y & copy "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.doc.ps1 & copy "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.pdf.ps1 & copy "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.rtf.ps1 & copy "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.xls.vbs & copy "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.png.vbs & copy "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.doc.ps1 & copy "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.pdf.ps1 & copy "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File %temp%\T1036.007_masquerading.rtf.ps1		"2280"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
2026-07-27 16:35:02.855	C:\Windows\System32\whoami.exe		"C:\Windows\system32\whoami.exe"		"2280"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
2026-07-27 16:35:02.848	C:\Windows\System32\HOSTNAME.EXE		"C:\Windows\system32\HOSTNAME.EXE"		"2280"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
2026-07-27 16:35:17.147	C:\Windows\System32\HOSTNAME.EXE		"C:\Windows\system32\HOSTNAME.EXE"		"2280"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading_rtf.ps1
2026-07-27 16:35:17.289	C:\Windows\System32\HOSTNAME.EXE		"C:\Windows\system32\HOSTNAME.EXE"		"2280"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File C:\Users\longday\AppData\Local\Temp\T1036.007_masquerading_rtf.ps1

Dùng powershell mở cmd

"cmd.exe" /c

Copy đổi tên calc thành file có đuôi các dạng file tài liệu docx, pdf, powershell

copy "C:\Windows\System32\calc.exe" %temp%\T1036.007_masquerading.docx.exe /Y &

copy "C:\Windows\System32\calc.exe" %temp%\T1036.007_masquerading.pdf.exe /Y &

copy "C:\Windows\System32\calc.exe" %temp%\T1036.007_masquerading.ps1.exe /Y &

Copy file có sẵn trong AtomicRedTeam ra thư mục temp là thư mục có quyền đọc ghi rồi đổi tên thành các file như xls, xlsx, png, doc, pdf

```
copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.vbs"
%%temp%%\T1036.007_masquerading.xls.vbs /Y &

copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.vbs"
%%temp%%\T1036.007_masquerading.xlsx.vbs /Y &

copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.vbs"
%%temp%%\T1036.007_masquerading.png.vbs /Y &

copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.ps1"
%%temp%%\T1036.007_masquerading.doc.ps1 /Y &

copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.ps1"
%%temp%%\T1036.007_masquerading.pdf.ps1 /Y &

copy "C:\AtomicRedTeam\atomics\T1036.007\src\T1036.007_masquerading.ps1"
%%temp%%\T1036.007_masquerading.rtf.ps1 /Y & %%temp%%\T1036.007_masquerading.docx.exe
& %%temp%%\T1036.007_masquerading.pdf.exe &
%%temp%%\T1036.007_masquerading.ps1.exe & %%temp%%\T1036.007_masquerading.xls.vbs
& %%temp%%\T1036.007_masquerading.xlsx.vbs &
%%temp%%\T1036.007_masquerading.png.vbs &
```

Sau khi copy xong thì dùng powershell để thực thi lần lượt các file đuôi .ps1

```
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File
%%temp%%\T1036.007_masquerading.doc.ps1 &
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File
%%temp%%\T1036.007_masquerading.pdf.ps1 &
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -File
%%temp%%\T1036.007_masquerading.rtf.ps1
```

7. MITRE ATT&CK Mapping

Thành phần	Ảnh xạ
Tactic hiện tại	Stealth
Technique	T1036 - Masquerading
Sub-technique	T1036.007 - Double File Extension
Atomic Test	Test #1 - File Extension Masquerading
Procedure	Đặt phần mở rộng tài liệu/hình ảnh trước phần mở rộng thực thi thật
Execution Methods	EXE, WScript/VBScript và PowerShell
Detection Evidence	File creation, process execution, command line và parent-child process

8. Impact and Risk Assessment

Trong môi trường thực tế, kỹ thuật này có thể:

- Đánh lừa người dùng mở file tưởng là Word, PDF, Excel hoặc hình ảnh.
- Che giấu loại file thực sự như .exe, .vbs, .ps1 hoặc .lnk.
- Cho phép thực thi mã từ email attachment hoặc thư mục người dùng có quyền ghi.
- Làm tăng khả năng thành công của phishing và initial access.
- Hỗ trợ malware thực thi trước khi người dùng nhận ra loại file thật.
-

9. Final Conclusion

LimaCharlie ghi nhận đầy đủ chuỗi tạo file và thực thi tiến trình; Splunk tổng hợp thành cảnh báo Critical - Risk Score 90 và chuyển thành công sang DFIR-IRIS.

Kết luận	Giá trị
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Final Verdict	True Positive - Authorized Simulation
Execution Result	Successful
Detection Result	Successful
Prevention Result	Requires Validation
Security Risk	Critical if unauthorized
Business Impact	None
IOC Identified	No
Containment	Not Required
Cleanup	Must be verified
Case Status	Close after cleanup and prevention validation